A SURJECTIVE ETALE CHART GIVES AN ETALE EQUIVALENCE RELATION WITH A COEQUALIZER DIAGRAM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement contains an xymatrix display; it is part of the counted statement span and is preserved verbatim, with the chapter's own Xy-pic setup retained in the preamble.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Definition of an equivalence relation on a category. Groupoids, Definition groupoids-definition-equivalence-relation, used at proof lines 1393-1395. Supplies the relation axioms checked on T-valued points.
- Definition of injective and surjective maps of sheaves. Sites, Definition sites-definition-sheaves-injective-surjective, used at proof lines 1417-1419. Is the sense in which U -> F is shown surjective.
- Surjectivity implies the coequalizer statement. Sites, Lemma sites-lemma-coequalizer-surjection, used at proof lines 1424-1426. Concludes the coequalizer diagram.

Source reference. Original definition 022P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2714–2714. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 00WM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1188–1188. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 086K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10395–10395. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let F be an algebraic space over S. Let $f : U \to F$ be a surjective étale morphism from a scheme to F. Set $R = U \times_F U$. Then*

(1) *$j : R \to U \times_S U$ defines an equivalence relation on U over S (see Groupoids, Definition 022P).*
(2) *the morphisms $s, t : R \to U$ are étale, and*

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(3) *the diagram*

$$R \rightrightarrows U \longrightarrow F$$

is a coequalizer diagram in $Sh((Sch/S)_{fppf})$.

Proof. Let T/S be an object of $(Sch/S)_{fppf}$. Then $R(T) = \{(a, b) \in U(T) \times U(T) \mid f \circ a = f \circ b\}$ which defines an equivalence relation on $U(T)$. The morphisms $s, t : R \to U$ are étale because the morphism $U \to F$ is étale.

To prove (3) we first show that $U \to F$ is a surjection of sheaves, see Sites, Definition 00WM. Let $\xi \in F(T)$ with T as above. Let $V = T \times_{\xi, F, f} U$. By assumption V is a scheme and $V \to T$ is surjective étale. Hence $\{V \to T\}$ is a covering for the fppf topology. Since $\xi|_V$ factors through U by construction we conclude $U \to F$ is surjective. Surjectivity implies that F is the coequalizer of the diagram by Sites, Lemma 086K. □